

Notes: Cooper and John (QJE, 1988)

Coordinating Coordination Failures in Keynesian Models

Contents

1	One-sentence summary	3
2	Big picture	3
2.1	The paper's central question	3
2.2	What the paper unifies	3
2.3	Core mechanism	3
2.4	Main takeaway	3
3	Incremental contribution of the paper	3
3.1	A unified framework for coordination failures	3
3.2	Separating spillovers from strategic complementarities	3
3.3	Formal link between multiplicity and complementarity	4
3.4	Formal link between multipliers and complementarity	4
3.5	Connection to fix-price Keynesian models	4
3.6	Why the contribution is durable	4
4	Conceptual framework	4
4.1	The abstract game	4
4.2	Spillovers	4
4.3	Strategic complementarities	4
4.4	Coordination failure	5
4.5	Multiplier effects	5
5	Model setup and derivations	5
5.1	Best responses and the reaction function	5
5.2	Multiple equilibria	5
5.3	Efficiency and spillovers	5
5.4	Pareto-ranked equilibria	5
5.5	Multiplier condition	5
6	Main propositions	6
6.1	Proposition 1: complementarity and multiplicity	6
6.2	Proposition 2: spillovers and inefficiency	6
6.3	Proposition 3: positive spillovers and insufficient activity	6
6.4	Proposition 4: Pareto ranking of equilibria	6
6.5	Proposition 5: continuum of equilibria	6
6.6	Proposition 6: complementarity and multipliers	6

7	Economic examples	6
7.1	Production-function complementarities	6
7.2	Matching technologies	6
7.3	Demand spillovers	7
7.4	Fix-price models	7
8	Identification and interpretation	7
8.1	What the paper identifies theoretically	7
8.2	Why the framework is not just a metaphor	7
8.3	Limits of the theory	7
8.4	Policy interpretation	7
9	Tables and figures	7
9.1	Figure I: Reaction function and multiple symmetric Nash equilibria; Figure II: Continuum of equilibria	7
9.2	Figure III: Sector reaction curves and the Keynesian cross	8
9.3	Notes-created Table: Main propositions and mechanisms	9
10	Short summary	9
11	Conclusion	10
11.1	Core conclusion	10
11.2	Why the paper matters	10
11.3	Incremental contribution revisited	10
11.4	Policy implication	10
11.5	Limitations	10
11.6	Takeaway	10

1 One-sentence summary

Cooper and John show that many Keynesian coordination-failure models share one common logic: **strategic complementarities** create multiple or amplified equilibria, while **spillovers** make those equilibria inefficient and often Pareto-ranked.

2 Big picture

2.1 The paper’s central question

The paper asks what makes a macroeconomic model “Keynesian” when the model does not rely on nonrational expectations, fixed wages, or fixed prices. Its answer is that Keynesian features can arise from the interaction of strategic complementarities and spillovers.

2.2 What the paper unifies

Cooper and John argue that seemingly different models—search models, increasing-returns models, matching models, demand-spillover models, and fix-price models—share the same underlying structure. Agents’ actions affect other agents’ payoffs and also affect the marginal return to other agents’ actions.

2.3 Core mechanism

Strategic complementarity means that when others raise their action, my best response rises too:

$$V_{12}(e_i, \bar{e}) > 0.$$

This creates upward-sloping reaction functions. If the reaction function crosses the 45-degree line more than once, the economy has multiple symmetric Nash equilibria. If positive spillovers are present, higher-action equilibria are often socially preferred.

2.4 Main takeaway

The paper’s contribution is conceptual rather than empirical: it gives a general language for understanding why decentralized economies may get stuck at low-output equilibria and why shocks can be multiplied through strategic interactions.

3 Incremental contribution of the paper

3.1 A unified framework for coordination failures

Earlier papers such as Diamond, Bryant, Hart, and Weitzman used different microfoundations for coordination failures. Cooper and John show that their common element is strategic complementarity. This makes coordination failure a general property of games, not a curiosity of one special model.

3.2 Separating spillovers from strategic complementarities

The paper distinguishes two ideas that are often blurred. **Spillovers** concern how one agent’s action affects another agent’s payoff. **Strategic complementarities** concern how one agent’s action affects another agent’s marginal payoff from acting. Spillovers create inefficiency; complementarities create multiplicity and multiplier effects.

3.3 Formal link between multiplicity and complementarity

The paper proves that strategic complementarity is necessary for multiple symmetric Nash equilibria. This turns the intuitive Keynesian idea of mutually reinforcing actions into a formal condition on the cross partial derivative of payoffs.

3.4 Formal link between multipliers and complementarity

The paper proves that strategic complementarity is necessary and sufficient for multiplier effects in the simple symmetric game. A shock to one agent has an amplified aggregate effect because the shock shifts others' best responses.

3.5 Connection to fix-price Keynesian models

The paper shows that fixed-price economies can also be interpreted as strategic-complementarity environments. Quantity constraints in one market affect other markets, generating the same multiplier logic found in Keynesian cross diagrams.

3.6 Why the contribution is durable

The paper gives later macroeconomics a vocabulary—strategic complementarity, spillovers, Pareto-ranked equilibria, and coordination failure—that can be applied across models of aggregate demand, search, finance, networks, technology adoption, and organizational complementarities.

4 Conceptual framework

4.1 The abstract game

There are I agents. Agent i chooses a scalar action e_i . The paper focuses on symmetric situations and writes the payoff of agent i as

$$V(e_i, \bar{e}),$$

where $\bar{e}$ summarizes the actions of all other agents. A symmetric Nash equilibrium (SNE) is an e such that

$$e_i = e \quad \text{and} \quad e \in \arg \max_{e_i} V(e_i, e).$$

4.2 Spillovers

Spillovers appear when the payoff of one agent changes directly with the actions of others:

$$V_2(e_i, \bar{e}) \neq 0.$$

Positive spillovers mean that when other agents raise their action, my payoff rises. In macro examples, positive spillovers often arise because another agent's production, search, or demand expands my opportunities.

4.3 Strategic complementarities

Strategic complementarity appears when another agent's action raises my marginal return to my own action:

$$V_{12}(e_i, \bar{e}) > 0.$$

This implies an upward-sloping best response. When others invest, produce, search, or demand more, it becomes optimal for me to do more as well.

4.4 Coordination failure

A coordination failure occurs when the economy is in a Nash equilibrium that is Pareto dominated by another feasible equilibrium. No individual wants to deviate alone, but everyone would prefer a coordinated move to the higher-action equilibrium.

4.5 Multiplier effects

A multiplier effect occurs when a primitive shock has an aggregate equilibrium effect larger than the direct individual response. In the paper's notation, the multiplier is driven by the slope of the reaction function. If the slope is positive and below one, individual responses feed back into one another but remain locally stable.

5 Model setup and derivations

5.1 Best responses and the reaction function

Let $e_i^*(\bar{e})$ be agent i 's best response to the actions of all other agents. Under standard concavity conditions,

$$\frac{de_i^*(\bar{e})}{d\bar{e}} = -\frac{V_{12}}{V_{11}}.$$

Since $V_{11} < 0$ at an interior maximum, strategic complementarity $V_{12} > 0$ implies an upward-sloping best response.

5.2 Multiple equilibria

Symmetric Nash equilibria are intersections of the reaction function with the 45-degree line:

$$e_i^*(e) = e.$$

Multiple crossings require that the reaction function be upward-sloping in some range. Hence strategic complementarity is necessary for multiple SNE.

5.3 Efficiency and spillovers

A symmetric competitive equilibrium maximizes individual payoffs holding others' actions fixed. A symmetric cooperative equilibrium internalizes the effect of one agent's action on others. With spillovers,

$$V_2(e_i, \bar{e}) \neq 0,$$

private and social incentives diverge. Positive spillovers imply that low-action equilibria tend to involve too little action.

5.4 Pareto-ranked equilibria

If there are positive spillovers globally, equilibrium payoffs rise as the common action rises. Thus multiple SNE can be Pareto-ranked by the level of activity: higher-action equilibria dominate lower-action equilibria.

5.5 Multiplier condition

Let p denote the slope of the reaction function around a stable SNE. The equilibrium multiplier has the form

$$\frac{1}{1-p},$$

so a positive multiplier larger than one requires $p > 0$. Since $p > 0$ is strategic complementarity, strategic complementarity is necessary and sufficient for the multiplier effect in the paper's game.

6 Main propositions

6.1 Proposition 1: complementarity and multiplicity

Strategic complementarity is necessary for multiple symmetric Nash equilibria. The graphical intuition is Figure I: multiple intersections with the 45-degree line require a positively sloped reaction function.

6.2 Proposition 2: spillovers and inefficiency

If the game exhibits spillovers at a symmetric Nash equilibrium, the equilibrium is inefficient relative to the cooperative benchmark. Agents do not internalize their effect on the payoffs of others.

6.3 Proposition 3: positive spillovers and insufficient activity

With positive spillovers, any low-action equilibrium is locally dominated by nearby higher-action allocations. The problem is not too much private initiative but too little coordinated action.

6.4 Proposition 4: Pareto ranking of equilibria

When multiple SNE exist and spillovers are positive globally, higher-action equilibria Pareto dominate lower-action equilibria. This is the paper's formal version of a Keynesian coordination failure.

6.5 Proposition 5: continuum of equilibria

If the reaction curve coincides with the 45-degree line over an interval, and spillovers are positive, there is a continuum of Pareto-ranked equilibria over that interval.

6.6 Proposition 6: complementarity and multipliers

Strategic complementarity is necessary and sufficient for multiplier effects in the abstract game. This connects the mathematical cross-partial condition to the Keynesian multiplier idea.

7 Economic examples

7.1 Production-function complementarities

The paper first studies complementarities arising inside a production process. If one agent's input raises the marginal product of another agent's input, then individual actions become strategic complements. Coordination failures can occur because each producer waits for others to produce more.

7.2 Matching technologies

The paper also connects its framework to search and matching models. If more search effort by others raises the value of my own search or production decision, then search activity can display strategic complementarities. This connects the Cooper-John framework to Diamond-style unemployment models.

7.3 Demand spillovers

In the imperfect-competition example, firms in different sectors have demands that depend on income earned in other sectors. When sector 2 produces more, it creates income that supports demand for sector 1. This is the core mechanism behind the Keynesian cross in Figure III.

7.4 Fix-price models

The paper finally shows that fix-price models can be interpreted through the same lens. If a firm in one sector produces more, it relaxes demand constraints in other sectors. The same strategic complementarity behind flexible-price examples also operates in fix-price economies.

8 Identification and interpretation

8.1 What the paper identifies theoretically

The paper identifies the common structural condition behind a class of Keynesian outcomes. It does not estimate parameters. Instead, it shows that multiple equilibria, multipliers, and coordination failures can all be traced to strategic complementarities combined with spillovers.

8.2 Why the framework is not just a metaphor

The abstract game yields precise propositions. Once a model can be mapped into this game, the existence of spillovers and complementarities immediately tells us what kinds of macroeconomic phenomena may arise: inefficiency, Pareto-ranked equilibria, and multiplier effects.

8.3 Limits of the theory

Strategic complementarity is necessary for multiple equilibria in the paper's simple game, but it is not sufficient. The reaction curve must be sufficiently steep. The paper also abstracts from dynamic adjustment, equilibrium selection, heterogeneous agents, and policy implementation.

8.4 Policy interpretation

If the economy is stuck in a low-action equilibrium, a coordinated policy intervention can move agents toward a better equilibrium. The model does not say exactly which policy is best, but it explains why policies that coordinate expectations or raise aggregate demand can have large effects.

9 Tables and figures

9.1 Figure I: Reaction function and multiple symmetric Nash equilibria; Figure II: Continuum of equilibria

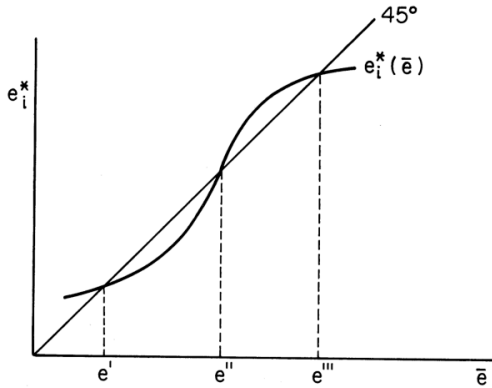
Read:

- Figure I shows an upward-sloping reaction function crossing the 45-degree line three times.
- The three intersections are symmetric Nash equilibria.
- The low and high intersections are stable if the reaction slope is less than one locally; the middle intersection is unstable in the usual graphical sense.

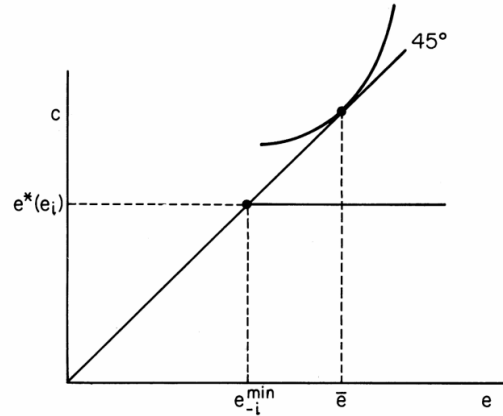
- Figure II shows a flat best-response segment overlapping with the 45-degree line, generating a continuum of equilibria.

Economic message: Figure I is the graphical core of the paper: strategic complementarity makes multiple equilibria possible. Figure II shows the limiting case where a whole interval of equilibria exists, and positive spillovers make higher points in the interval better.

Detailed interpretation: These two figures formalize Keynesian coordination failure without fixed prices or irrational expectations. Agents are optimizing, but because the return to my action depends on what others do, the economy can settle at a low-action equilibrium even when everyone would prefer a coordinated move to a high-action equilibrium.



(a) **Figure I: Reaction function.** The S-shaped best response crosses the 45-degree line multiple times, illustrating how strategic complementarity can create multiple symmetric Nash equilibria.



(b) **Figure II: Continuum of equilibria.** A flat segment overlapping the 45-degree line generates a continuum of equilibria; positive spillovers rank higher-action equilibria above lower-action equilibria.

9.2 Figure III: Sector reaction curves and the Keynesian cross

Read:

- Figure III plots sectoral reaction curves in a two-sector imperfect-competition economy.
- The curve for sector 1 is $Q_1 = \bar{M}/P^* + (1/2)Q_2$.
- The curve for sector 2 is symmetric: $Q_2 = \bar{M}/P^* + (1/2)Q_1$.
- Their intersection determines equilibrium output Q^* .
- The slope is the spending feedback from one sector's income into the other sector's demand.

Economic message: Figure III connects the abstract game to the Keynesian cross. Sectoral output decisions are strategic complements because production in one sector creates demand for the other sector. An autonomous demand shock shifts one reaction curve and is amplified through the feedback loop.

Detailed interpretation: The figure shows why the multiplier equals more than the direct effect. If sector 1 expands, income rises and demand shifts toward sector 2. Sector 2 then expands, feeding demand back to sector 1. The model therefore produces a Keynesian multiplier from optimizing firms in imperfect competition, not from exogenous price rigidity alone.

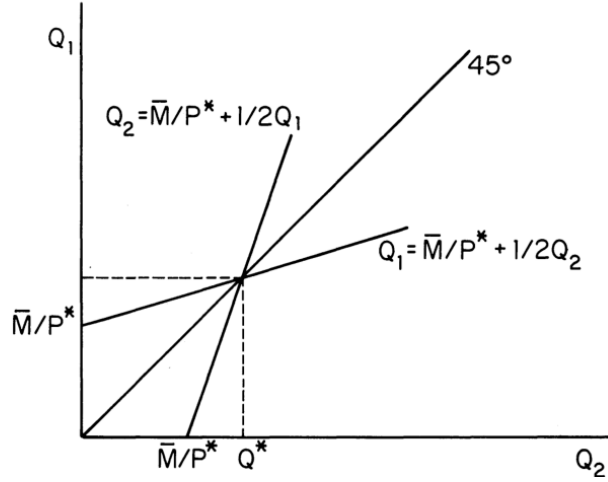


Figure III: Sector reaction curves and Keynesian cross. This original figure shows how output in one sector raises demand for output in the other sector, generating a multiplier process. As requested, the single unpaired figure is shown at width 0.6.

9.3 Notes-created Table: Main propositions and mechanisms

Result	Formal content	Economic meaning
Prop. 1	Strategic complementarity is necessary for multiple SNE.	Multiple equilibria require upward-sloping best responses.
Prop. 2	Spillovers imply SNE inefficiency.	Agents ignore their effects on other agents' payoffs.
Prop. 3	Positive spillovers imply higher nearby actions can improve welfare.	Low-action equilibria feature too little activity.
Prop. 4	Multiple SNE with positive spillovers are Pareto-ranked.	High-output equilibria dominate low-output equilibria.
Prop. 5	A flat reaction segment can create a continuum of equilibria.	Coordination can select any point on an equilibrium interval.
Prop. 6	Strategic complementarity is necessary and sufficient for multipliers.	Positive feedback produces aggregate amplification.

Notes-created table: Main propositions and mechanisms. This table is not an original table from the paper. It summarizes the theoretical results and their economic meanings.

10 Short summary

- The paper provides a general framework for Keynesian coordination failures.
- Spillovers and strategic complementarities are distinct: spillovers create inefficiency, while complementarities create multiple equilibria and multipliers.
- Multiple symmetric Nash equilibria require upward-sloping best responses.
- With positive spillovers, higher-action equilibria Pareto dominate lower-action equilibria.
- Strategic complementarity is necessary and sufficient for multiplier effects in the paper's abstract game.
- Production, matching, demand, and fix-price models can all generate strategic complementarities.

- The paper explains Keynesian outcomes without assuming irrational expectations or exogenous wage-price rigidity.

11 Conclusion

11.1 Core conclusion

Cooper and John show that strategic complementarities are the common engine behind a broad class of Keynesian models. When actions are mutually reinforcing, the economy can have multiple equilibria and multiplier effects. When spillovers are positive, low-action equilibria are inefficient and Pareto dominated by high-action equilibria.

11.2 Why the paper matters

The paper transforms the discussion of Keynesian coordination failure from a collection of examples into a general theory. It shows that the important question is not whether a model has search, increasing returns, matching, imperfect competition, or fixed prices. The important question is whether the model has positive cross-effects in strategies and payoffs.

11.3 Incremental contribution revisited

The paper's incremental contribution is to identify the minimal conceptual ingredients behind coordination failures: strategic complementarity for multiplicity and multipliers, spillovers for welfare failure. This provides a modular way to build and compare macroeconomic models with Keynesian features.

11.4 Policy implication

If an economy is in a low-output equilibrium, policy can potentially coordinate agents on a better equilibrium. The role of policy is not only to fix prices or provide information; it may also coordinate beliefs, raise aggregate demand, or shift reaction curves so that private incentives support the high-action equilibrium.

11.5 Limitations

The model is deliberately abstract. It does not solve equilibrium selection, dynamic adjustment, heterogeneity, empirical measurement, or the political economy of coordination policy. It also shows necessary conditions, not a universal sufficient condition for multiplicity in every possible model.

11.6 Takeaway

The enduring lesson is that macroeconomic outcomes may be bad because agents fail to coordinate, not because they fail to optimize. Strategic complementarities make private actions mutually reinforcing; spillovers make the resulting equilibria socially ranked. Together they provide a simple grammar for Keynesian coordination failures.